

# "Dual-prudential" regulation and monetary policy transmission

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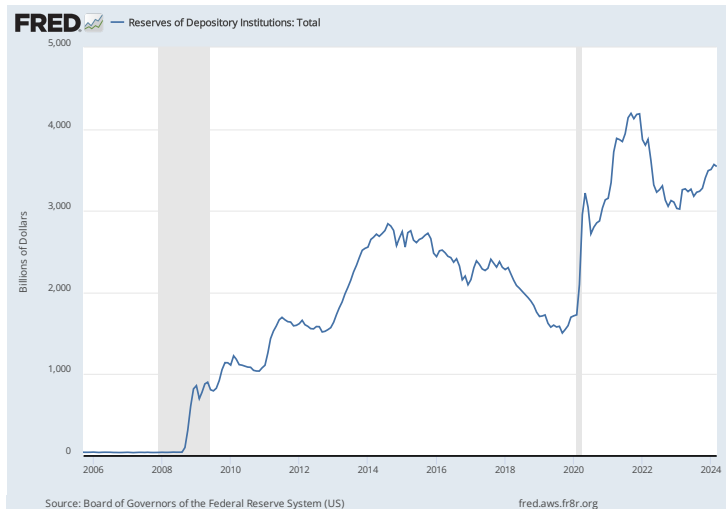
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# Overview

1. Motivation
2. Empirical (anecdotal) evidence
3. Model (in progress)

# We are now in the ample reserves regime



# Motivation

- **Question:** How do post-GFC macro-prudential regulations affect
  - ▶ Commercial banks' asset/liability/liquidity management?
  - ▶ Conventional and unconventional monetary policy transmission?
- Macro-prudential regulations require banks to hold more liquid assets (reserves and treasuries) based on internal stress testing results.
  - ▶ Are banks constrained by liquidity regulations?
  - ▶ Are the excess reserves enough to meet new liquidity demand?
- What is the transmission mechanism of QE/QT?
  - ▶ Credit channel: commercial banks' balance sheet expansion.
  - ▶ Liquidity channel: reduce liquidity premium so as to lower borrowing cost.

# This project

- A framework with banking liquidity regulations to analyze monetary policy transmission.
- Liquidity regulations such as liquidity coverage ratio (LCR) and stress testing (LST) requirements post dual constraints on liquidity holdings.
  - ▶ Liquidity demand = stress state outflow + intraday temporary outflow.
  - ▶ Reserves are financed by demand deposit, which simultaneously add to stress state outflow and intraday temporary outflow.
  - ▶ As a result, no effective excess reserves.
- Depending on the size of the precautionary reserve holdings for intraday flow risk, QE can affect bank's financing cost in opposite directions.

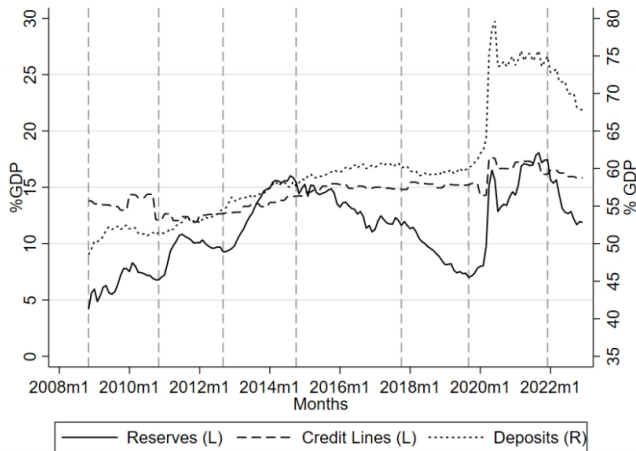
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# Bank finance reserves with demand deposits (Archaya and Rajan, 2024 JF)

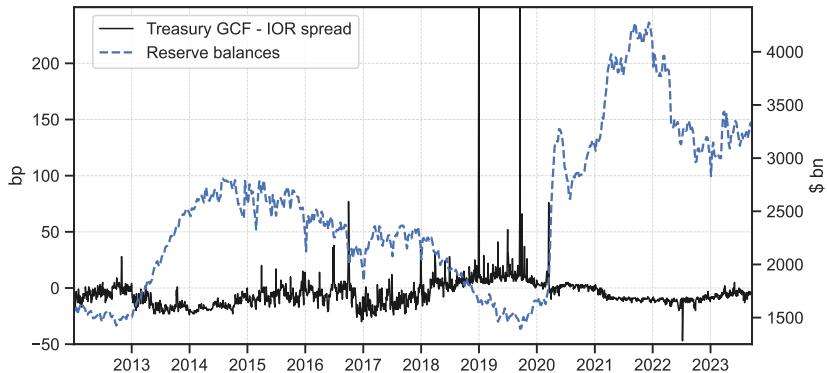
- Demand deposits are uninsured and subject to run

Panel A. Credit Lines, Deposits, and Reserves as percentage of GDP



# Repo market spikes in Sept 2019

- One explanation: reserves flew to the treasury account.



## Repo market spikes in Sept 2019

- The chairman and CEO of JPMorgan Chase mentioned compliance with liquidity regulations as one reason of not lending on the FFM.
- *As I said, we have \$120 bn in our checking account at the Fed, and it goes down to \$60 bn and then back to \$120 bn during the average day. But we believe the requirement under CLAR (Comprehensive Liquidity Analysis and Review) and resolution and recovery is that we need enough in that account, so if there's extreme stress during the course of the day, it doesn't go below zero. If you go back to before the crisis, you'd go below zero all the time during the day. So the question is, how hard is that as a red line? Was the intent of regulators between CLAR and resolution to lock up that much of reserves in the account with Fed? And that'll be up to regulators to decide. But right now, we have to meet those rules and we don't want to violate anything we've told them we're going to do.*

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# Bank

- Bank's balance sheet:
  - ▶ Asset: reserves  $R$ , loans  $A$ .
  - ▶ Liabilities: uninsured demand deposit  $D^d$ , insured deposit  $D^i$ .
- Intra-day flow risk:  $\epsilon \sim U([-D^d\sigma, D^d\sigma])$ .
  - ▶ Proportional to demand deposit.
  - ▶ With friction-less FFM, Aggregate flow.
  - ▶ Idiosyncratic flow risk with matching friction in the FFM.
- LCR constraint must hold for the worst case scenario:
  - ▶  $\theta D^d$  is the outflow in stress state over a 90 days' period.
  - ▶ The LCR itself only requires that  $R \geq \theta D^d$ .
  - ▶ However, due to intraday liquidity risk, LCR+LST post a dual constraint.

$$(R - \theta D^d) \geq \sigma D^d$$

# Bank

- Maximize profits

$$\begin{aligned} \max_{R, A, D^d, D^i} \quad & R(1 + r^r) + A(1 + r^a) - g(A/e) - (1 + r^f)f - (1 + r^d)D^d - (1 + r^i)D^i \\ \text{s.t.} \quad & A + R - D^d - D^i - f = e, \quad R \geq (\theta + \sigma)D^d \end{aligned}$$

- $f$  is borrowing from the Federal Fund Market.
- $g$  represents a leverage cost or risk premium.

# Household

- Asset: demand deposit, insured deposit, risky assets
- Liability: future labor tax
- Household has a liquidity preference  $U(D^d)$ .
- Maximize utility

$$\max_{D^i, D^d} D^i(1 + r^i) + D^d(1 + r^d) + U(D^d), \quad D^i + D^d = W$$

- When the central bank changes reserve supply,  $W$  changes one for one.
  - ▶ Incomplete market: household can not pay future taxes today.

# Partial equilibrium

- An equilibrium is (i) a set of interest rate  $\{r^a, r^r, r^d, r^i\}$  (ii) a set of portfolio choices  $\{D_b^i, D_b^d, A, f, D_h^d, D_h^i\}$ , such that given prices, the portfolio choices solve bank and household's problem and market clears

$$R = \bar{R}, \quad f = 0, \quad D_b^i = D_h^i, \quad D_b^d = D_h^d$$

## First order conditions

- Risk adjusted loan return equals to borrowing cost.

$$r^a - g'(A/e)/e = r^i$$

- IOR  $r^r$  is exogenously given. The FFR is determined by the shadow value

$$r^f = r^r + 1/(\theta + \sigma)(r^i - r^d)$$

- The quantity of demand deposit is constrained by reserves.

$$R = (\theta + \sigma)D^d$$

- The liquidity premium is determined by marginal utility in liquidity.

$$u'(D^d) = r^i - r^d$$

# Quantitative easing

- Suppose in the equilibrium

$$r^d < r^r < r^i$$

- The central bank increases reserves by  $\Delta R$ .
- If  $\sigma + \theta > 1$ , then  $\Delta D^d < \Delta R$ , meaning bank has to rely more on insured-deposit financing  $\Rightarrow$  higher financing cost.
- If  $\sigma + \theta < 1$ , then  $\Delta D^d > \Delta R$ , meaning bank use less insured-deposit financing and thus has lower financing cost.
- "The deposit channel of QE"
- Next steps: Model how bank's financing cost affect the broader economy  $\Rightarrow$  need a dynamic model.